PAPER_690: Fornax Galaxy Cluster: N-Body Gravitational Dynamics and UQFF

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Abstract

The Fornax Cluster (Abell S373) is the second-richest nearby galaxy cluster, 19.5 Mpc distant, dominated by the cD galaxy NGC 1399. This paper models cluster dynamics via N-body simulation with UQFF gravitational corrections.

1. Cluster Parameters

Parameter	Value
$M_{cluster}$	$7 \times 10^{13} M_{\odot}$
R_{virial}	1 Mpc
Distance	19.5 Mpc
Members (bright)	58

2. UQFF Cluster Gravity

$$g_{UQFF}(r) = \frac{GM_{cluster}}{r^2}\left(1 + \frac{\rho_{SCm}}{\rho_{UA}}\right)(1 + f_{TRZ})$$

3. Velocity Dispersion (Virial Theorem)

$$\sigma_v = \sqrt{\frac{GM_{cluster}}{2R_{virial}}} \approx 370 \text{ km/s}$$

4. Tidal Radius

$$r_{tidal} = r_{orbit}\left(\frac{m_{gal}}{3M_{cluster}}\right)^{1/3}$$

5. N-Body Equations of Motion

$$\ddot{\mathbf{r}}_i = \sum_{j \neq i} \frac{Gm_j(\mathbf{r}_j - \mathbf{r}_i)}{|\mathbf{r}_{ij}|^3 + \epsilon^3}$$

References

- Drinkwater et al. (2001), MNRAS, 326, 1076
- UQFF Framework, Star-Magic Session 174

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